Expt 5

Start ssh to connect WinSCP to the ubuntu

sudo apt install net-tools

sudo apt update

sudo apt install openssh-server -y

sudo systemctl enable ssh

sudo systemctl start ssh

sudo systemctl status ssh

sudo ufw status

Initialize git repo from ubuntu terminal

cd ~/Desktop/myproject

git init

git status

git add .

git commit -m "Initial commit"

git branch -M main

Connect Local repo to Github

git remote add origin https://github.com/USERNAME/REPOSITORY.git

git push -u origin main

Command	Purpose
sudo apt install net-tools	Installs the Net-tools package, which provides networking commands such as ifconfig and netstat.
sudo apt update	Updates the package list to get information about the latest available software packages.
sudo apt install openssh-server -y	Installs the OpenSSH server to allow secure remote access to the system using SSH.
sudo systemctl enable ssh	Configures the SSH service to start automatically when the system boots.
sudo systemctl start ssh	Starts the SSH service immediately.
sudo systemctl status ssh	Displays the current status of the SSH service, such as active or inactive.
sudo ufw status	Checks whether the UFW firewall is active or inactive and displays its rules.

Command	Purpose
cd ~/Desktop/myproject	Navigates to the project directory.
git init	Initializes a new Git repository.
git status	Checks the current status of files in the repository.
git add .	Adds all project files to the staging area.
git commit -m "Initial commit"	Saves the staged changes with a commit message.
git branch -M main	Renames the current branch to main.
git remote add origin https://github.com/USERNAME/REPOSITORY.git	Connects the local repository to the GitHub repository.
git push -u origin main	Uploads the local commits to GitHub and sets main as the default upstream branch for future pushes.

Expt 6

On Host Machine

touch cloud_computing.html

git add .

git commit -m "ADDED cloud_computing.html file"

git push origin main

git remote set-url origin git@github.com:raghunandangaude24-commits/CC_LAB.git

ssh -T git@github.com

git push origin main

On VM

git pull origin main

Command	Purpose
ls	Lists files and folders in the current directory.
cd CC_LAB	Enters the CC_LAB directory.
touch cloud_computing.html	Creates an empty HTML file.
git add .	Adds all files to the staging area.
git commit -m "ADDED cloud_computing.html file"	Saves staged changes with a commit message.
git push origin main	Uploads local commits to the GitHub main branch.
git remote set-url origin git@github.com:raghunandangaude24-commits/CC_LAB.git	Changes the remote repository URL to SSH.
ssh -T git@github.com	Tests SSH authentication with GitHub.
git push origin main	Pushes the committed changes to GitHub using SSH.
git pull origin main	Fetches the latest changes from the GitHub main branch and merges them into the current local branch.

Expt 7

Make your friend clone your repo

cd CC

git clone

cd CC_LAB

touch EXPT-07

ls

nano EXPT-07

cat EXPT-07

git add .

git commit -m "New File Entry"

git push origin main

Command	Purpose
cd CC	Navigates into the CC directory.
git clone <repository-url>	Downloads and copies a remote GitHub repository to the local system.
cd CC_LAB	Enters the cloned CC_LAB repository directory.
touch EXPT-07	Creates a new empty file named EXPT-07.
ls	Lists all files and folders in the current directory.
nano EXPT-07	Opens the file in the Nano text editor to write or edit content.
cat EXPT-07	Displays the contents of the file in the terminal.
git add .	Adds all new and modified files to the staging area.
git commit -m "New File Entry"	Saves the staged changes with a commit message.
git push origin main	Uploads the committed changes to the GitHub main branch.

Change Port Number on WinSCP

sudo nano /etc/ssh/sshd_config

Change port number to 2222

```
# Note that the above implies that configuration options from snippets in
# /etc/ssh/sshd_config.d/*.conf take precedence over those defined in this
# file. In addition, configuration snippet files are processed in lexical
# order, so options defined in files with names that sort earlier take
# precedence.
#
# Such configuration snippets may be present in default installations of
# Ubuntu.
Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
```

Run this command to restart the server

sudo nano /etc/ssh/sshd_config

sudo systemctl daemon-reload

sudo systemctl restart ssh.socket

sudo systemctl restart ssh

Command	Purpose
sudo nano /etc/ssh/sshd_config	Opens the SSH server configuration file in Nano editor with administrator privileges to modify SSH settings.
sudo systemctl daemon-reload	Reloads the systemd manager configuration after changes to service or socket unit files.
sudo systemctl restart ssh.socket	Restarts the SSH socket service to apply socket-related configuration changes.
sudo systemctl restart ssh	Restarts the SSH server service to apply the updated SSH configuration.

Expt 8

Create a index.html file on the VM



Create a new repo and push this file in the github from the VM

Deploy it on the vercel

Install and start MySQL server

sudo apt install mysql-server

Login to mysql and create table

sudo mysql

```
CREATE DATABASE cc_db;
USE cc_db;

CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100),
  contact VARCHAR(15),
  address VARCHAR(255)
);

INSERT INTO users (name, contact, address)
VALUES ('Raghunandan Gaude', '9999999999', 'Goa, India');

SELECT * FROM users;
```

Enable Remote Connection

sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf

```
#
# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address            = 0.0.0.0
mysqlx-bind-address     = 127.0.0.1
#
```

Create a remote user

CREATE USER ‘remoteuser’@‘%’ INDENTIFIED BY ‘raghunandan’;

GRANT ALL PRIVILEGES ON test.* TO ‘remoteuser’@‘%’;

FLUSH PRIVILEGES;

Remote DB access via workbench

GRANT ALL PRIVILEGES ON *.* TO ‘remoteuser’@‘%’ WITH GRANT OPTION;

Command	Purpose
sudo apt install mysql-server	Installs MySQL Server on Ubuntu.
sudo systemctl start mysql	Starts the MySQL server service.
sudo mysql	Logs in to the MySQL command-line interface as administrator.
CREATE DATABASE cc_db;	Creates a database named cc_db.
USE cc_db;	Selects the cc_db database.
CREATE TABLE users (...);	Creates a table named users with id, name, contact and address columns.
INT AUTO_INCREMENT	Automatically generates an increasing integer ID.
PRIMARY KEY	Uniquely identifies each record in the table.
VARCHAR(100)	Stores text up to 100 characters.
INSERT INTO users (...) VALUES (...);	Inserts user details into the table.
SELECT * FROM users;	Displays all records from the users table.
sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf	Opens the MySQL configuration file to enable remote connections.
bind-address = 0.0.0.0	Allows MySQL to listen for connections from other network devices.
CREATE USER 'remoteuser'@'%' IDENTIFIED BY 'raghunandan';	Creates a remote MySQL user who can connect from any host.
GRANT ALL PRIVILEGES ON test.* TO 'remoteuser'@'%';	Grants all privileges on the test database to the remote user.
FLUSH PRIVILEGES;	Reloads privilege changes in MySQL.

GRANT ALL PRIVILEGES ON *.* TO 'remoteuser'@'%' WITH GRANT OPTION;	Grants all privileges on all databases and permission to grant privileges to other users.
MySQL Workbench	Connects to the remote MySQL server and manages databases graphically.

Expt 9

Installation of MYSQL php and Apache:

sudo apt install apache2 php libapache2-mod-php mysql-server php-mysql -y

Start mysql

sudo mysql

Create database

```
CREATE DATABASE web_project;
USE web_project;

CREATE TABLE contacts(
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(50),
  email VARCHAR(100),
  message TEXT
);
```

Create php file to connect backend to frontend

sudo mkdir -p /var/www/html/contact-form

sudo chown -R $USER:$USER /var/www/html/contact-form

cd /var/www/html/contact-form

nano submit.php

```php
<?php

$conn = new mysqli("localhost", "remoteuser", "raghunandan", "web_project");

if($conn->connect_error) {
   die("Connection failed: " . $conn->connect_error);
}

$name = $_POST['name'];
$email = $_POST['email'];
$message = $_POST['message'];

$sql = "INSERT INTO contacts (name, email, message)
        VALUES ('$name', '$email', '$message')";

if ($conn->query($sql) === TRUE) {
    echo "Message submitted successfully";
} else {
    echo "Error: " . $conn->error;
}

$conn->close();

?>
```

nano index.html

```html
<!DOCTYPE html>
<html>
<head>
    <title>Contact Form</title>
</head>

<body>

    <h1>Contact Us</h1>

    <form action="https://showroom-parakeet-species.ngrok-free.dev/contact-form/submit.php" method="POST">

        <label>Name:</label><br>
        <input type="text" name="name" required><br><br>

        <label>Email:</label><br>
        <input type="email" name="email" required><br><br>

        <label>Message:</label><br>
        <textarea name="message" required></textarea><br><br>

        <button type="submit">Submit</button>

    </form>

</body>
</html>
```

Create new dir for ngrok server

mkdir ~/ngrok-setup

Installation of ngrok server

cd ~/ngrok-setup

pwd

wget https://bin.equinox.io/c/bNyj1mQVY4c/ngrok-v3-stable-linux-amd64.tgz

Checking installation bt starting the server

tar -xvzf ngrok-v3-stable-linux-amd64.tgz

ls

sudo mv ngrok /usr/local/bin

Sign Up or sign in to get Auth Key (Before running ngrok http 80)

```
Sign up for an account: https://dashboard.ngrok.com/signup
Get your credential: https://dashboard.ngrok.com/get-started/your-authtoken
```

ngrok config add-authtoken <your auth token>

ngrok http 80

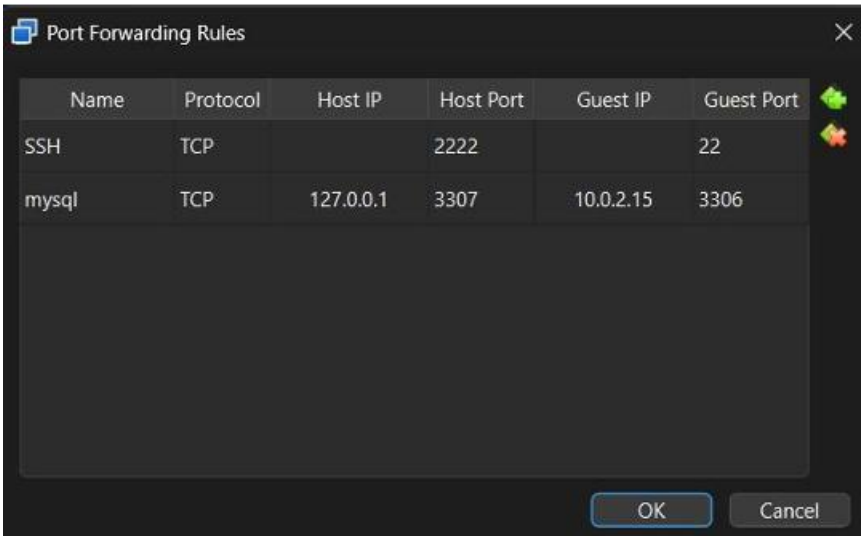
Deploy it on vercel

Command	Purpose
sudo apt install apache2 php libapache2-mod-php mysql-server php-mysql -y	Installs Apache web server, PHP, MySQL server and PHP-MySQL connectivity support.
sudo mysql	Opens the MySQL command-line interface with administrator privileges.
CREATE DATABASE web_project;	Creates a new database named web_project.
USE web_project;	Selects the web_project database.
CREATE TABLE contacts (id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(50), email VARCHAR(100), message TEXT);	Creates a contacts table to store ID, name, email and message.
sudo mkdir -p /var/www/html/contact-form	Creates a contact-form directory inside the Apache web server directory.
sudo chown -R $USER:$USER /var/www/html/contact-form	Changes ownership of the contact-form directory to the current user.
cd /var/www/html/contact-form	Navigates into the contact-form directory.
nano submit.php	Creates or opens the PHP file to write backend code for processing form submissions.
nano index.html	Creates or opens the HTML file to design the frontend contact form.

mkdir ~/ngrok-setup	Creates a new directory named ngrok-setup in the home directory.
cd ~/ngrok-setup	Navigates into the ngrok setup directory.
pwd	Displays the current working directory path.
wget https://bin.equinox.io/c/bNyj1mQVY4c/ngrok-v3-stable-linux-amd64.tgz	Downloads the Ngrok installation archive for 64-bit Linux.
tar -xvzf ngrok-v3-stable-linux-amd64.tgz	Extracts the downloaded Ngrok archive.
ls	Lists files and folders in the current directory.
sudo mv ngrok /usr/local/bin	Moves the Ngrok executable into /usr/local/bin to make it accessible from any directory.
ngrok config add-authtoken <your auth token>	Adds your Ngrok authentication token to configure your account.
ngrok http 80	Creates a public URL that forwards internet traffic to the local Apache server running on port 80.

Expt 10

Use this port forwarding in Oracle Virtual Box for Ubuntu



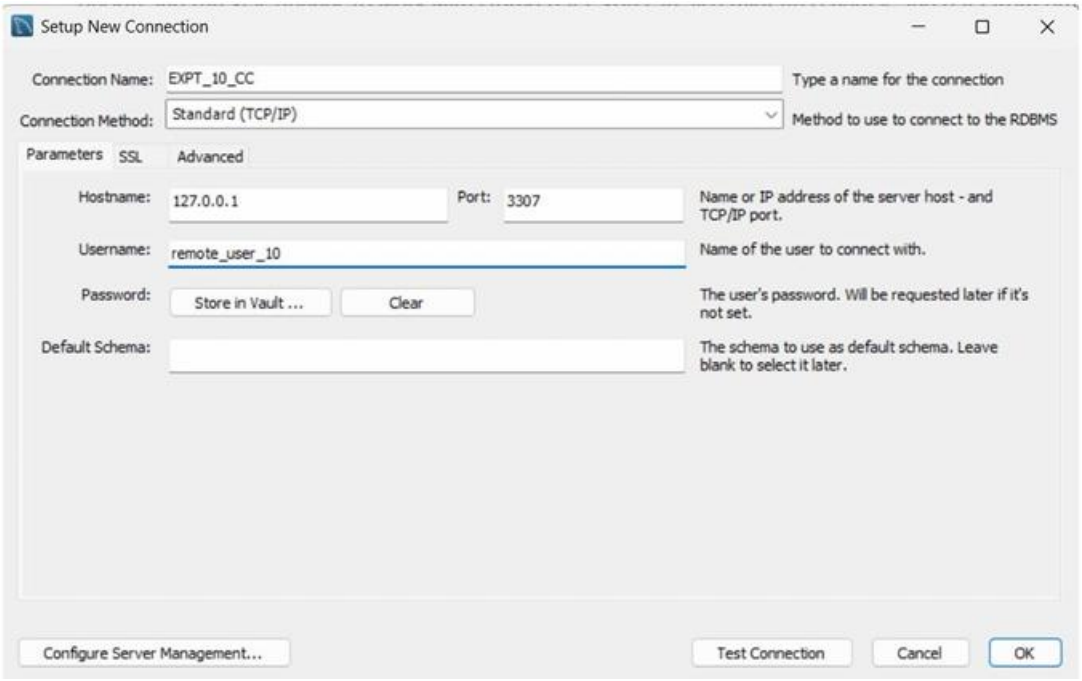
User previous configuration or create new configuration for remote access

```
CREATE USER 'remote_user_10'@'%' IDENTIFIED BY 'raghunandan';

GRANT ALL PRIVILEGES ON web_project.* TO 'remote_user_10'@'%';

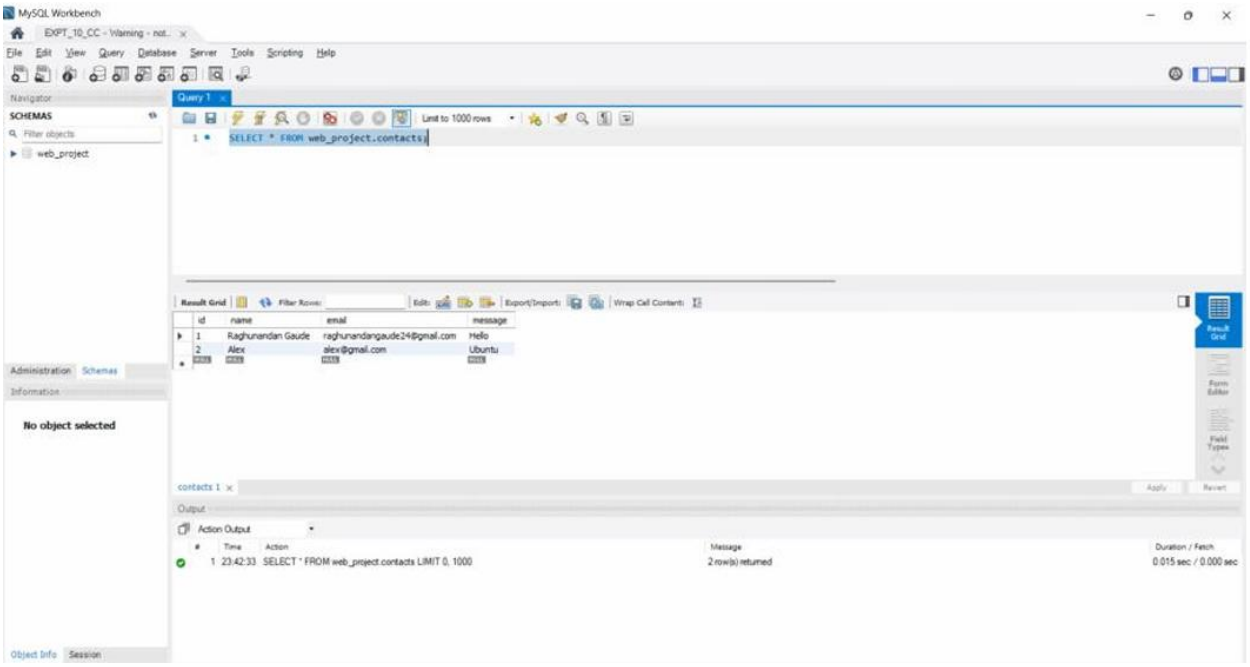
FLUSH PRIVILEGES;
```

Connect remote user in the workbench



Test Connection

Once connected, run queries to fetch data from server



CLOUD VS LOCAL VM VS HOST DB.

Environment	Description	Example	Pros	Cons
Local VM	MySQL runs on your own VirtualBox instance	Ubuntu/Kali VM	Full control, isolated testing	Needs local setup
Host DB	MySQL runs directly on your OS	XAMPP / WAMP	Fast access	Less isolation
Cloud DB	Hosted remotely	AWS RDS, Azure, GCP	Accessible anywhere, scalable	Costs, network dependency

Command	Purpose
CREATE USER 'remote_user_10'@'%' IDENTIFIED BY 'raghunandan';	Creates a new MySQL user named remote_user_10 with the specified password and allows connections from any host.
GRANT ALL PRIVILEGES ON web_project.* TO 'remote_user_10'@'%';	Grants all database privileges on the web_project database to the remote user.
FLUSH PRIVILEGES;	Reloads the privilege tables so that the updated permissions take effect.